

Songyang Zhang

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Education

University of Rochester

Ph.D. in Computer Science, Advisor: Jiebo Luo

Rochester, NY

Aug 2018 – Jul 2023

Zhejiang University

M.E. in Computer Science and Technology, Advisor: Jun Xiao

Hangzhou, China

Sep 2015 – Mar 2018

Southeast University

B.E. in Computer Science and Technology

Nanjing, China

Aug 2011 – Jun 2015

Professional Experience

Amazon

Applied Scientist at Amazon AGI

Santa Clara, CA

Jul 2023 – Present

- 2025 - video generation model scale up and omni model integration
 - * Refactored our video generation codebase to support more parallelism strategies.
 - * Led and optimized I2V training and inference pipeline.
 - * Led and built our in-house high-compression autoencoder.
 - * Led and built instruction-tuning pipeline to align MLLM with diffusion models.
- 2024 - Amazon Nova Reel 1.0 (launched at re:Invent 2024)
 - * Developed the initial codebase for video generation from scratch.
 - * Designed and built UNet-style and DiT-style model architecture.
 - * Led and optimized the model parallel strategies, training recipe and inference pipeline.
- 2023 - Amazon Titan Image Generator (launched at re:Invent 2023)
 - * Built a novice-friendly T2I personalization pipeline.

Meta

Part-Time Research Intern at Fundamental AI Research (FAIR) Accel, Mentor: Xi Yin

Menlo Park, CA

May 2022 - Aug 2022

- Research on Text-to-Video Generation (ICLR 2023, 1 Submission)

Meta

Research Intern at Fundamental AI Research (FAIR) Accel, Mentor: Xi Yin

Menlo Park, CA

Jan 2022 - Apr 2022

- Research on Text-to-Video Generation (ECCV 2022)

Tencent

Part-time Research Intern at Tencent AI Lab, Mentor: Linfeng Song, Lifeng Jin, Kun Xu

Bellevue, WA

Sep 2021 - Dec 2021

- Research on Unsupervised Grammar Induction (EMNLP 2022)

Microsoft

Research Intern at at Microsoft Azure, Mentor: Quanzeng You, Jiang Wang

Redmond, WA

May 2021 - Aug 2021

- Research on Moment Localization with Natural Language

Tencent

Research Intern at Tencent AI Lab, Mentor: Linfeng Song, Lifeng Jin, Kun Xu

Bellevue, WA

May 2020 - Aug 2020

- Research on Unsupervised Grammar Induction (NAACL 2021)
- Best Long Paper Award in NAACL 2021 (1/1797)

Microsoft

Research Intern at Microsoft Research Asia (MSRA), Mentor: Houwen Peng, Jianlong Fu

Beijing, China

May 2019 - Aug 2019

- Research on Moment Localization with Natural Language (AAAI 2020, TPAMI 2021)
- Rank 1 in HACS Temporal Action Localization Challenge at ICCV 2019

Tencent

Research Intern at Tencent AI Lab, Mentor: Linchao Bao, Lin Ma

Shenzhen, China

Mar 2018 - Aug 2018

- Research on Temporal Action Localization

Publications

* *Equal contribution, ordered alphabetically*

Ali Vosoughi, Shijian Deng, **Songyang Zhang**, Yapeng Tian, Chenliang Xu and Jiebo Luo, “Possible Worlds VQA: Cross Modality Bias Reduction in Visual Question Answering Systems from a Causal View.” *TMM*, 2024

Jie An*, **Songyang Zhang***, Harry Yang, Sonal Gupta, Jia-Bin Huang, Jiebo Luo, Xi Yin, “Latent-Shift: Latent Diffusion with Temporal Shift for Efficient Text-to-Video Generation.” *Submitted*, 2023

Uriel Singer, Adam Polyak, Thomas Hayes, Xi Yin, Jie An, **Songyang Zhang**, Qiyuan Hu, Harry Yang, Oron Ashual, Oran Gafni, Devi Parikh, Sonal Gupta, Yaniv Taigman, “Make-A-Video: Text-to-Video Generation without Text-Video Data.” *ICLR* 2023

Songyang Zhang, Linfeng Song, Lifeng Jin, Kun Xu, Dong Yu and Jiebo Luo, “Learning a Grammar Inducer from Massive Uncurated Instructional Videos.” *EMNLP* 2022 [Oral]

Xingchen Li, Long Chen, Jian Shao, Shaoning Xiao, **Songyang Zhang**, Jun Xiao, “Rethinking the Evaluation of Unbiased Scene Graph Generation.” *BMVC* 2022 [Oral]

Bolin Ni, Houwen Peng, Minghao Chen, **Songyang Zhang**, Gaofeng Meng, Jianlong Fu, Shiming Xiang, Haibin Ling, “Expanding Language-Image Pretrained Models for General Video Recognition.” *ECCV* 2022 [Oral]

Thomas Hayes*, **Songyang Zhang***, Xi Yin, Guan Pang, Sasha Sheng, Harry Yang, Songwei Ge, Isabelle Hu, Devi Parikh, “MUGEN: A Playground for Video-Audio-Text Multimodal Understanding and GENeration.” *ECCV* 2022

Lin Li, Long Chen, Yifeng Huang, Zhimeng Zhang, **Songyang Zhang**, Jun Xiao, “The Devil is in the Labels: Noisy Label Correction for Robust Scene Graph Generation.” *CVPR* 2022 [Oral]

Jiahui Li, Kun Kuang, Lin Li, Long Chen, **Songyang Zhang**, Jian Shao, Jun Xiao, “Instance-wise or Class-wise? A Tale of Neighbor Shapley for Concept-based Explanation.” *ACM MM* 2021

Zhengyuan Yang, **Songyang Zhang**, Liwei Wang, Jiebo Luo, “SAT: 2D Semantics Assisted Training for 3D Visual Grounding.” *ICCV* 2021 [Oral]

Songyang Zhang, Linfeng Song, Lifeng Jin, Dong Yu, Jiebo Luo, “Video-aided Unsupervised Grammar Induction.” *NAACL* 2021 [Best Long Paper Award]

Songyang Zhang, Houwen Peng, Jianlong Fu, Yijuan Lu, Jiebo Luo, “Multi-Scale 2D Temporal Adjacency Networks for Moment Localization with Natural Language.” *TPAMI* 2021

Shaoning Xiao, Long Chen, **Songyang Zhang**, Wei Ji, Jian Shao, Lu Ye, Jun Xiao, “Boundary Proposal Network for Two-Stage Natural Language Video Localization.” *AAAI* 2021

Songyang Zhang, Tolga Aktas, Jiebo Luo, “Mi YouTube es Su YouTube? Analyzing the Cultures using YouTube Thumbnails of Popular Videos.” *IEEE Big Data* 2021

Li Sun*, Haoqi Zhang*, **Songyang Zhang**, Jiebo Luo, “Content-based Analysis of the Cultural Differences between TikTok and Douyin.” *IEEE Big Data* 2020

Jie An, Tianlang Chen, **Songyang Zhang**, Jiebo Luo, “Global Image Sentiment Transfer.” *ICPR* 2020

Songyang Zhang, Houwen Peng, Le Yang, Jianlong Fu, Jiebo Luo, “Learning 2D Temporal Adjacent Networks for Temporal Action Localization.” HACS Temporal Action Localization Challenge at *ICCV* 2019 [Winner]

Songyang Zhang, Houwen Peng, Jianlong Fu, Jiebo Luo, “Learning 2D Temporal Adjacent Networks for Moment Localization with Natural Language.” *AAAI* 2020

Songyang Zhang, Jinsong Su, Jiebo Luo, “Exploiting Temporal Relationships in Video Moment Localization with Natural Language.” *ACMMM* 2019

Jiageng Feng, **Songyang Zhang**, Jun Xiao, “Explorations of skeleton features for LSTM-based action recognition.” *MTA* 2017

Songyang Zhang, Jun Xiao, Xiaoming Liu, Yang Yi, Yueting Zhuang, “Fusing Geometric Features for Skeleton-Based Action Recognition using Multilayer LSTM Networks.” *TMM* 2017

Songyang Zhang, Xiaoming Liu, Jun Xiao, “On Geometric Features for Skeleton-Based Action Recognition Using Multilayer LSTM Networks.” *WACV* 2017

Patents

Sonal Gupta, Adam Polyak, Thomas Falstad Hayes, Xi Yin, Jie An, Chao Yang, Oron Ashual, Oran Gafni, Devi Niru Parikh, Yaniv Nechemia Taigman, Uriel Singer, **Songyang Zhang** and Qiyuan Hu, “Text-to-Video Generation” *U.S. Patent 18/477,887, Filed Sep 29 2023. Patent Pending.*

Honors & Awards

Best Long Paper Award at NAACL 2021	<i>Jun 2021</i>
Winner, HACS Temporal Action Localization Challenge at ICCV 2019	<i>Oct 2019</i>
Student Travel Award at ACM Multimedia 2019	<i>Oct 2019</i>
Winner, RoboCup China Open 2014 Competition	<i>Oct 2014</i>

Services

Reviewer:

- Conference: ICIG’21, ACMMM’21, CVPR’22, BMVC’22, ACMMM’22, ACL APR’22, ACL’23, NeuIPS’23, ICLR’24
- Journal: TPAMI’18, TII’19, TMM’20, NEUCOM’20, TIP’20, TCSVT’21, TMM’21, IJCV’22, TPAMI’23